

10 Application of Nuclear Magnetic Resonance Imaging to Study Preferential Water Flow Through Root Channels

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ABSTRACT

Two nuclear magnetic resonance (NMR) imaging techniques, spin echo and gradient recalled echo (GRE) with radio frequency (RF) spoiling, were used to characterize the spatial distribution of bush bean (*Phaseolus vulgaris* L.) and sunflower (*Helianthus annuus* L.) roots in plaster of paris columns, 73 mm in diam. and 70 mm in height. The degree of root decay as well as water movement in root channels were observed by sequential NMR imaging studies during 6 months. The average infiltration rate of porous matrix, and the volumetric discharge rate of permeating root channels were also measured during the decay period. High quality NMR images in live roots in situ were obtained with a three-dimensional resolution of 0.4 by 0.4 by 0.6 mm. Root decay reduced the signal to noise ratio of NMR because of dessication and tissue disintegration. Root channels became preferential flow paths after a 4-wk decay period. Dry dead roots were hydrophilic, and could be rehydrated within 5 to 10 min after wetting. Spatial characteristics of roots, such as size, permeation, tortuosity, and orientation of elongation, did not affect the occurrence of preferential flow. The preferential flow velocity approximated from the measured volumetric discharge and the estimated total cross sectional area of permeating root channels was 10 to 50 mm s⁻¹, about 10⁴ to 10⁵ times higher than the average infiltration rate of the porous matrix.

Growing plant roots modify soil structure actively. In forcing their way through soil, roots displace soil particles. Around the roots, these particles

are further aligned and compressed by root radial expansion, and then stabilized by mucigelaneous root exudates and residues (Barley, 1968; Greacen et al., 1968). These stable root channels have been identified as preferential flow paths from dye studies (Barley, 1954; Petersson et al., 1987; Mitchell et al., 1991), and the water conducting effectiveness of these channels was found to depend on the stage of root decay (Barley, 1954; Petersson et al., 1987; Meek et al., 1989; Beven, 1991; Mitchell et al., 1991). However, no studies to date have measured the three-dimensional spatial distribution of roots and its effect on preferential flow over time with root decay. The threshold stage of root decay at which root channels become preferential flow paths is also unclear. Our objective is to investigate the use of NMR imaging for obtaining information of the spatial and temporal characteristics of plant roots in situ, and their effects on preferential flow.

Nuclear magnetic resonance imaging was originally designed for noninvasive study of the human body for medical purposes (Pykett, 1982). This imaging technique probes the magnetic moment of nuclei. After nuclei in a static magnetic field are probed by a RF magnetic pulse, they release energy and produce a measurable RF signal. Spatial distribution of the nuclei can be encoded into the signal by applying magnetic field gradients (Pykett, 1982; Wehrli et al., 1988).

In the recent decade, NMR imaging has been applied to studying water and ion transport in intact plant tissues (van As & Schaafsma, 1984; Yazaki et al., 1988; Bottomley et al., 1986), anatomical examinations of roots and stems (Brown et al., 1986, 1988), and in situ imaging of root system (Omasa et al., 1985). Many studies have applied NMR spectroscopy and NMR imaging to the measurement of soil water content (Omasa et al., 1985; Tollner et al., 1987; Tollner & Rollwitz, 1988; Paetzold et al., 1987; Anderson & Gantzer, 1987). We exploit several proton NMR imaging methods for measuring the spatial distribution of roots in situ, monitoring root decay, and identifying the preferential flow paths. The NMR imaging methods applied include spin echo and GRE with RF spoiling.

NUCLEAR MAGNETIC RESONANCE IMAGING METHODS

Spin Echo

The Spin Echo sequence is currently the most commonly used imaging sequence (Wehrli et al., 1988). The spin echo is formed by applying sequentially a 90° RF pulse and a 180° RF pulse (Hahn, 1950; Fukeshima and Roeder, 1981; Wehrli et al., 1988; Pykett, 1982). A 90° RF pulse is an oscillating RF magnetic pulse that converts the net magnetization of nuclei to be 90° to the static field. For two-dimensional imaging the spatial distribution of nuclei is encoded by using a phase encoding gradient in between the 90° RF and the 180° RF pulses, and a frequency encoding gradient in an orthogonal direction with which the spin echo signal is sampled (Fig. 10-1). The RF

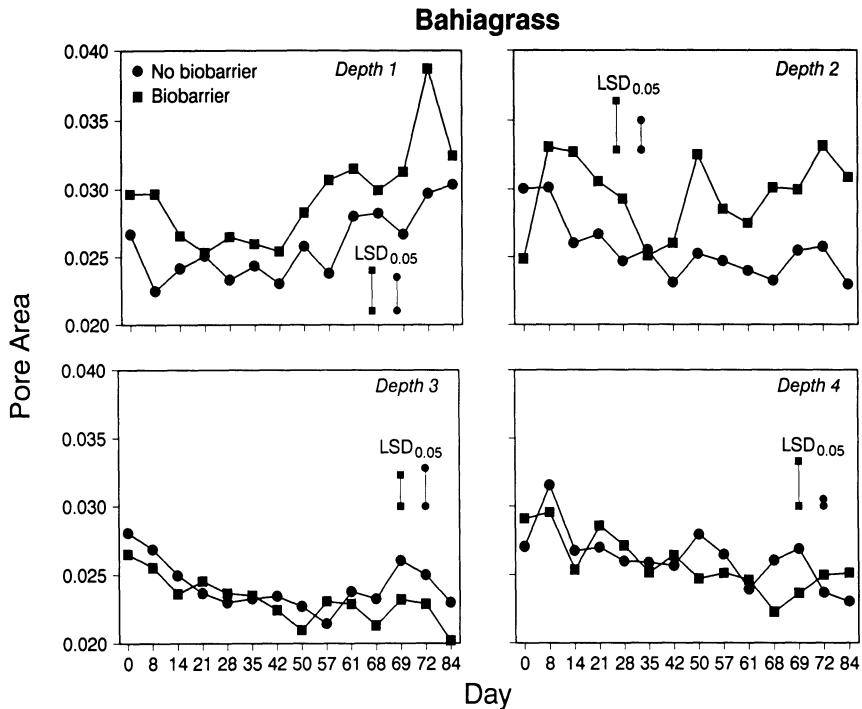


Fig. 10-1. The two-dimensional spin echo imaging sequence.

pulse can be applied in the presence of a magnetic field gradient to selectively excite a two-dimensional plane of a sample (Fig. 10-1), or it can be used as a nonselective pulse to obtain two-dimensional projection of a three-dimensional sample for the purpose of improving the NMR signal to noise ratio (SNR). A three-dimensional data set can also be obtained by using two orthogonal phase encoding gradients.

Nuclear magnetic resonance imaging is a weak phenomenon. Signal to noise ratio is always a problem (Wehrli et al., 1988). Proton images of soils generally show poor SNR and distortions because of the local magnetic field gradient generated on the surface of ferro-magnetic particles in soils (Rogers & Bottomley, 1987). Signal loss due to these particles have been the major obstacle of the application of NMR imaging to study soil systems (Rogers & Bottomley, 1987). Nonselective Spin Echo sequence and strong magnetic field gradients are two common strategies to minimize distortions caused by ferro-magnetic particles. The NMR signal strength is proportional to the size of imaging volume elements (voxels), however, the noise induced by the test sample is not (Wehrli et al., 1988). Increase in sampling volume by using a nonselective pulse sequence will therefore increase SNR. Since the resonance frequency is related to the sum of the static field and the gradient field (Wehrli et al., 1988), increase in gradient strength decreases the relative influence of field inhomogeneity.

Gradient Recalled Echo with Radio Frequency Spoiling

Although the conventional Spin Echo sequence described above produces satisfactory results, the process is relatively slow, especially for three-dimensional imaging, because a proper length of time is needed between the 90° and the 180° RF pulses, and between the data collection and the next 90° RF. In recent years many fast scan techniques, such as GRE and Fast spin echo (FSE), have been developed for clinical imaging (Zur et al., 1991; Constable et al., 1992a, b). The fast scan technique used in this study is a GRE sequence with RF spoiling, which speeds up the imaging process by using a small RF flip angle (smaller than 90°) and a gradient echo (Zur et al., 1991). The time between successive RF excitations (TR) is shorter than both the longitudinal and transverse relaxation parameters (T1 and T2 respectively) of the test sample. A phase shift that is incremented by changing and then resetting the frequency of RF synthesizer prior to every RF excitation is applied to eliminate or spoil the transverse magnetization (Zur et al., 1991). Since the longitudinal magnetization is not eliminated prior to RF pulses, the GRE sequence with RF spoiling is T1 dependent. Furthermore, this sequence is sensitive to field inhomogeneity and hence may not be applicable to soil systems with high content of ferro-magnetic particles.

PROCEDURES

Nuclear magnetic resonance imaging experiments were done in the Center for Functional Imaging, Lawrence Berkeley Laboratory. The imaging system consists of a 2.35 Tesla superconducting magnet (Bruker, Germany), and a home built spectrometer. The effects of the three-dimensional spatial distribution and the decay of roots on preferential water flow were studied by the sequential NMR imaging of roots and water flow in situ during a period of 24 wk. Five artificial roots in Hanford sandy loam were used in a preliminary test to examine the effect of ferro-magnetic particles on image quality. These artificial roots were made of plastic tubes filled with 0.01 M CuSO_4 solution. The diameters were 0.3, 0.4, 0.6, 0.7, and 0.8 mm. The smallest root was not detected. Distortion of NMR imaging of the roots in soil was observed even with the nonselective Spin Echo sequence and the maximum gradient strength of 3×10^{-2} Tesla m^{-1} along the read out direction (Fig. 10-2). In order to avoid artifacts caused by ferro-magnetic particles in soil, an artificial porous medium, plaster of paris, was used to simulate the soil matrix.

Bush bean and sunflower plants were grown in white sand (from Monterey, CA) packed in columns 73 mm in diameter and 90 mm in height. After a 25-d growth period, these plants were washed repeatedly until they were separated from the sand. Three samples were made by submerging the clean roots in a mixture of two parts of water and one part of plaster of paris, with the original root configuration maintained as much as possible. The cylindrical containers for the samples were 73 mm in diam. and 130 mm in

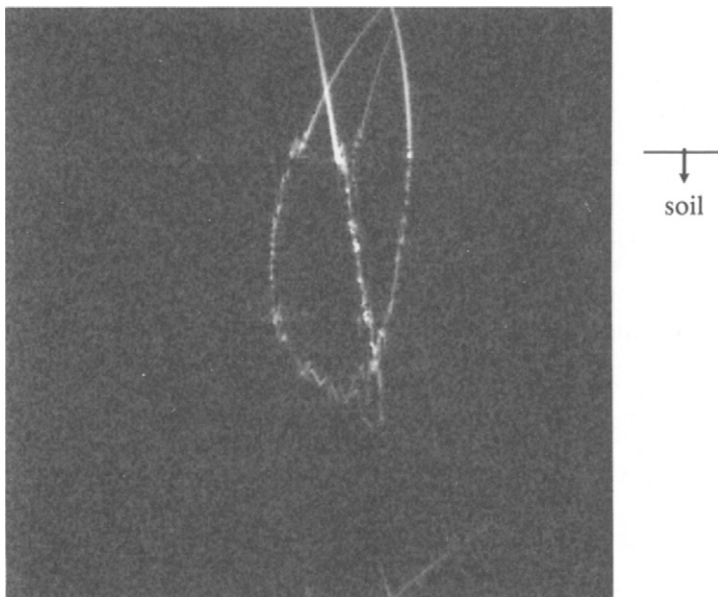


Fig. 10-2. Five artificial roots in Hanford sandy loam.

height. The height of the plaster of paris matrix was 70 to 90 mm. The bottom 20-mm section of each sample was cut off when the sample was fully hardened. The final heights of the containers and the plaster of paris matrices were 120 mm and 50 to 70 mm, respectively. The plant roots in the three samples were a complete bush bean root system, a complete sunflower root system, and six trimmed individual sunflower roots.

The hydraulic conductivities of the bulk samples were measured 1 wk after the plaster of paris samples were made, and 4 wk (for the bean sample) or 24 wk (for the rest of the samples) into the decay period. The three-dimensional GRE and the two-dimensional Spin Echo sequences were used to study the decaying roots in situ. The preferential flow paths were identified by taking NMR images of the samples during water penetration and after short wetting periods.

RESULTS

The spatial distribution of roots in plaster of paris was examined by the three-dimensional GRE imaging sequence 1 d after the samples were made. To examine the three-dimensional data sets, an image analysis program VIDA (Hoffman et al., 1992) was used to produce sequences of projections of the root systems, each rotated by a set angle from the previous projection. Movies of the rotating root systems can be generated by displaying the se-

quences in closed loops. Projections at two angles of each sample are shown in Fig. 10-3.

The sunflower roots were spread vertically down the sample matrix (Fig. 10-3a). Roots were concentrated mainly in two vertical regions near the sides of the plaster of paris column. The bush bean roots were spread mainly in the horizontal direction, and formed a dense mat in the upper 30 mm of the sample profile (Fig. 10-3b). Only two major vertical roots were observed in the bean sample. The third sample had six trimmed sunflower roots, and only two roots extended down to the bottom of the column (Fig. 10-3c). The voxel resolution of these images was 0.4 by 0.4 by 0.6 mm.

All the samples were kept in saturated condition during the first 2 wk of the decay period. The three-dimensional GRE images of the bean and sunflower samples after a 13-d decay period did not have significant changes. The image of the sample with six trimmed roots, however, was significantly weakened (Fig. 10-4, compared with Fig. 10-3c).

The samples were left to dry for 3 wk after the saturation period. After this dehydration period, NMR imaging could detect roots with weak signals only when the samples were rehydrated, and it became increasingly difficult to use the three-dimensional GRE sequence. Figure 10-5 showed the results of both the spin echo and the three-dimensional GRE imaging of the saturated samples 24 wk into the decay period. The Spin Echo sequence produced much better images than the three-dimensional GRE sequence.

The water flow paths in the samples with bean roots were studied after a 4 wk decay period. The average gravimetric water content of the sample before the experiment was 17%, and the spin echo imaging failed to detect any signal under such low water content. However, after water was applied to the sample surface, the spin echo imaging detected the flow paths in the sample (Fig. 10-6a). Water appeared in the bean root channels even though most of the channels were highly tortuous, horizontally spread, and not completely extended through the sample (nonpermeating). A cross sectional image of the stem 5 mm below the sample surface showed that water appeared at the center of the stem tissues (Fig. 10-6a), which indicated that dead plant tissues were hydrophilic.

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Nuclear magnetic resonance imaging of bean roots (5 wk into the decay process) was taken with the Spin Echo sequence before and after the dry bean sample (gravimetric water content = 20%) was wetted by a pulsed input of 100 mL of water for 5 min. Water discharged from the base of the

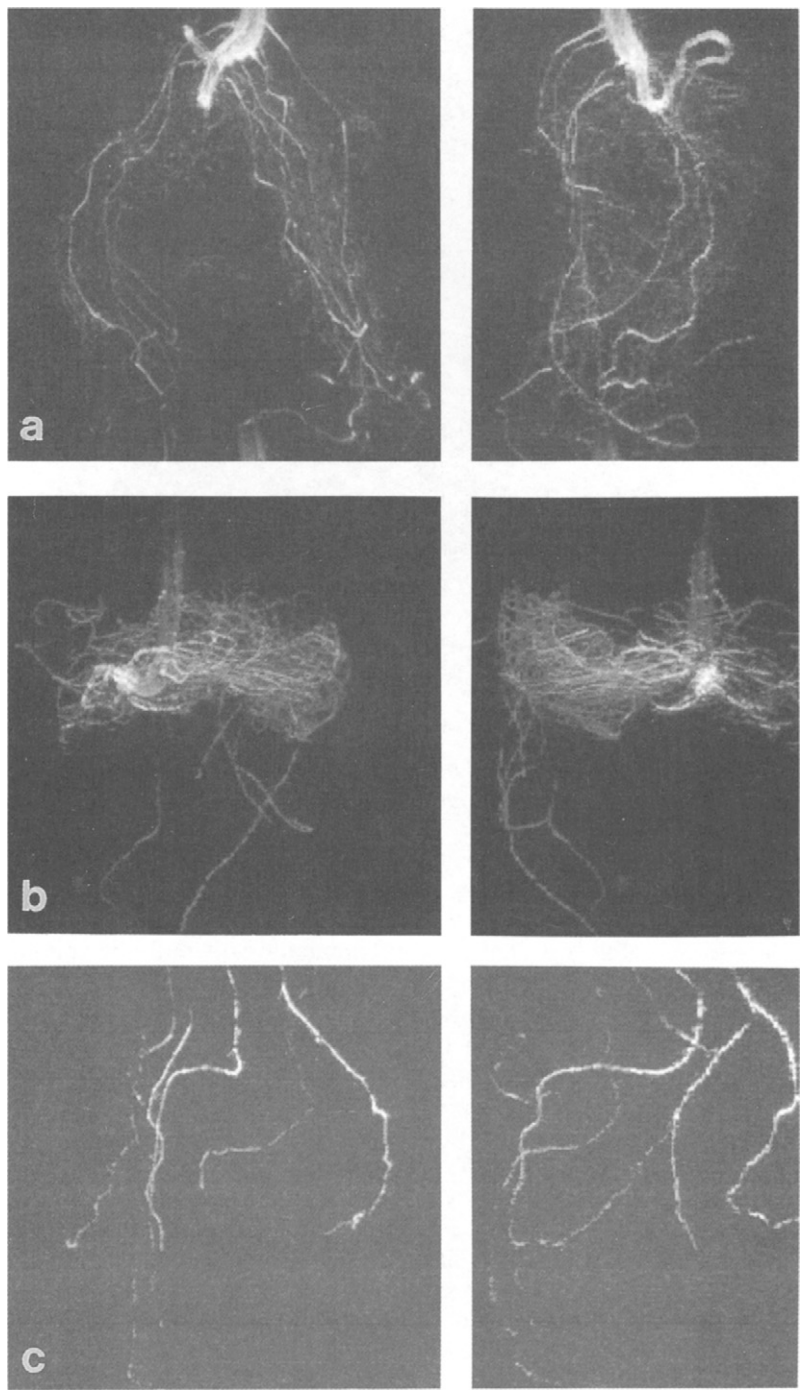


Fig. 10-3. Plant roots in plaster of paris 1 d after the samples were made. Each sample is viewed at two different angles. (a) sunflower roots, (b) bean roots, and (c) six trimmed sunflower roots.

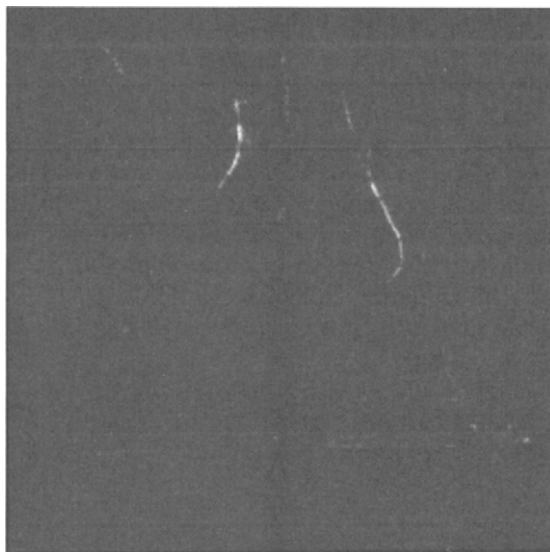


Fig. 10-4. Six trimmed sunflower roots 13 d into the decay process.

sample was collected with a filter paper, and the water left on the sample surface was removed after the 5-min wetting period. This wetting and imaging procedure was repeated once. All the water discharged in the two, 5-min wetting periods was from the only two channels reaching the bottom of the sample, indicating that flow was in the preferential flow paths created by roots.

The preferential flux density was estimated by dividing the volumetric discharge rate by the estimated total cross sectional area of the flow paths. The total cross sectional area of the preferential paths was calculated by subtracting the total cross sectional area of decaying roots from the total cross sectional area of channel openings at the sample bottom. The two channel openings at the sample bottom were 0.9 and 0.7 mm in diam. The decaying roots in the channels were 0.6 and 0.5 mm in diam. The estimated preferential flow was $4 \times 10^1 \text{ mm s}^{-1}$ (Table 10-1), which was 10^4 times faster than the average hydraulic conductivity measured at the same time, and 10^5 times higher than the average saturated hydraulic conductivity measured 1 wk after the bean sample was made (Table 10-1). No NMR image could be obtained before the sample was wetted. A faint image of the root mat appeared after the first 5-min wetting procedure (Fig. 10-7a). More roots were observed in the image taken after the second 5-min wetting procedure (Fig. 10-7b). The results from the other two samples, after being subjected to repeated 5-min wetting periods, were similar to those from the bean sample (Table 10-1; Fig. 10-7c, and Fig. 10-7d).

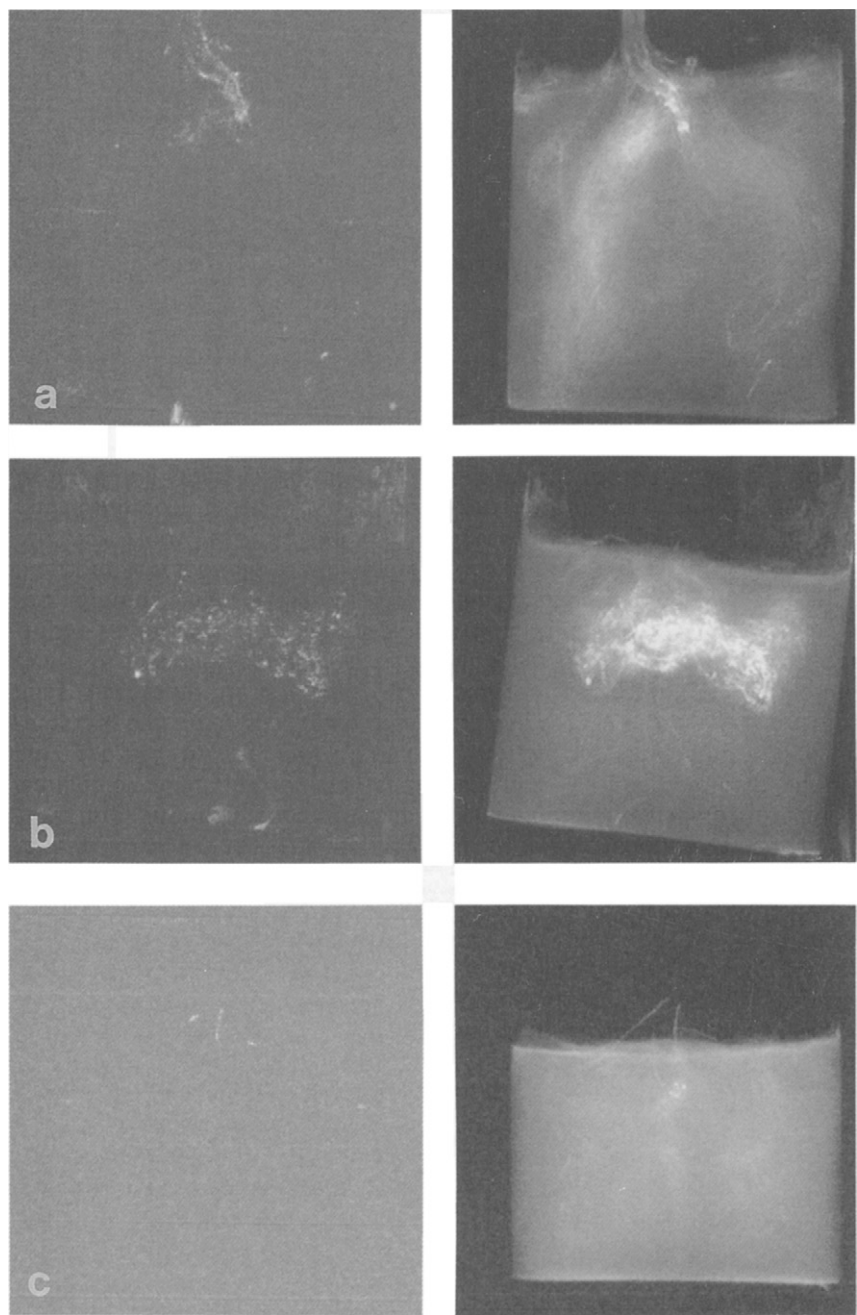


Fig. 10-5. Plant roots in plaster of paris 24 wk into the decay process. The pictures on the left are the three-dimensional Gradient Recalled Echo images, and those on the right are two-dimensional Spin Echo images. (a) sunflower roots, (b) bean roots, and (c) six trimmed sunflower roots.

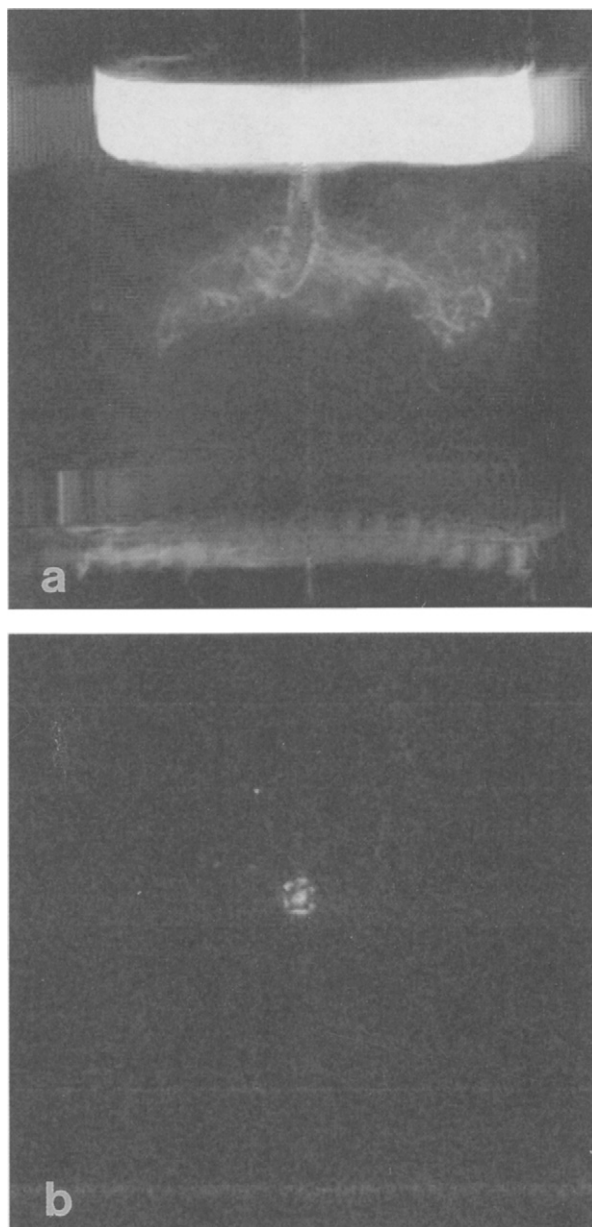


Fig. 10-6. (a) Bean roots in plaster of paris during water penetration. (b) The cross sectional view of the bean stem 5 mm below the sample surface, 6 h after water was applied.

DISCUSSION

Although the dead root material was hydrophilic, the SNR of NMR images in the rehydrated decaying roots was much lower than those of the live

Table 10-1. The water flow data.

	Bean	Sunflower	Six trimmed roots (STR)
$k_1 \dagger$, mm s ⁻¹	1×10^{-4}	2×10^{-4}	2×10^{-4}
$k_2 \ddagger$, mm s ⁻¹	5×10^{-3}	3×10^{-3}	7×10^{-4}
$v_p \S$, mm s ⁻¹	4×10^1	2×10^1	2×10^1

$\dagger k_1$, the average hydraulic conductivity, measured 1 wk after the samples were made. k_1 was calculated by dividing the discharge rate by the cross sectional area of the sample and the hydraulic gradient. Uncertainty: 20%.

$\ddagger k_2$, the average hydraulic conductivity, measured 4 wk into the decay period for the bean sample, and 24 wk into the decay period for the sunflower and the six trimmed root samples. Uncertainty: 20%.

$\S v_p$, the estimated preferential flux density. v_p was measured at the same time as k_2 . v_p was calculated by dividing the discharge rate by the estimated total cross sectional area of the permeating preferential flow paths. Uncertainty: 30%.

roots. The loss of SNR was most severe when the three-dimensional GRE sequence was used. It was possible that as roots decayed there was less tissue to adsorb water (Barley, 1954) and there were more air pockets in the tissue during rehydration. Consequently, the amount of water to be imaged was reduced, resulting in a lower SNR. The GRE sequence might have suffered additional signal loss due to an increase in field inhomogeneity at air-water interfaces. The GRE sequence is very sensitive to the T1 relaxation of the sample (Zur et al., 1991). The tap water used to rehydrate the decaying roots may have a longer T1 than the tissue water in live roots causing the poorer performance of the three-dimensional GRE sequence.

Previous studies showed that not all large pores conduct water (Bouma et al., 1977; Lindon & Dixon, 1976; Beven & Germann, 1982; Beven, 1991). The connectivity and continuity of pore structure in three-dimensional space may be more important than the size of pores (Bouma et al., 1977). Lindon and Dixon (1976) found that macropore flow was severely impeded when soil air pressure increased from 0 to 5×10^{-4} MPa. In this study most of the root channels in plaster of paris were nonpermeating (or dead-ended, Fig. 10-3). Since these channels were not connected to open space, water flow in channels of this type under a pulsed water input was expected to be impeded by encapsulated air. However, the NMR imaging of root channels in plaster of paris during water penetration (Fig. 10-6) and after short wetting periods (Fig. 10-7) showed that water moved into both permeating and nonpermeating channels filled with decaying roots during same wetting periods. Other spatial characteristics, such as the tortuosity and orientation of elongation, also had no apparent effect on the occurrence (or initiation) of water flow in root channels. Such results suggested that water flow in root channels may have been driven by capillary effect. The hydrophilic decaying root material may have behaved like a wick drawing water into the channels, and within each channel there may have been a fast flow region in the gap between the decaying roots and the porous matrix, and a slow diffusive flow region in decaying root tissue. The hypotheses of the wicking effect and flow regions within root channels will have to be tested with data of the gap size between the roots and matrix, the amount of roots remaining in chan-

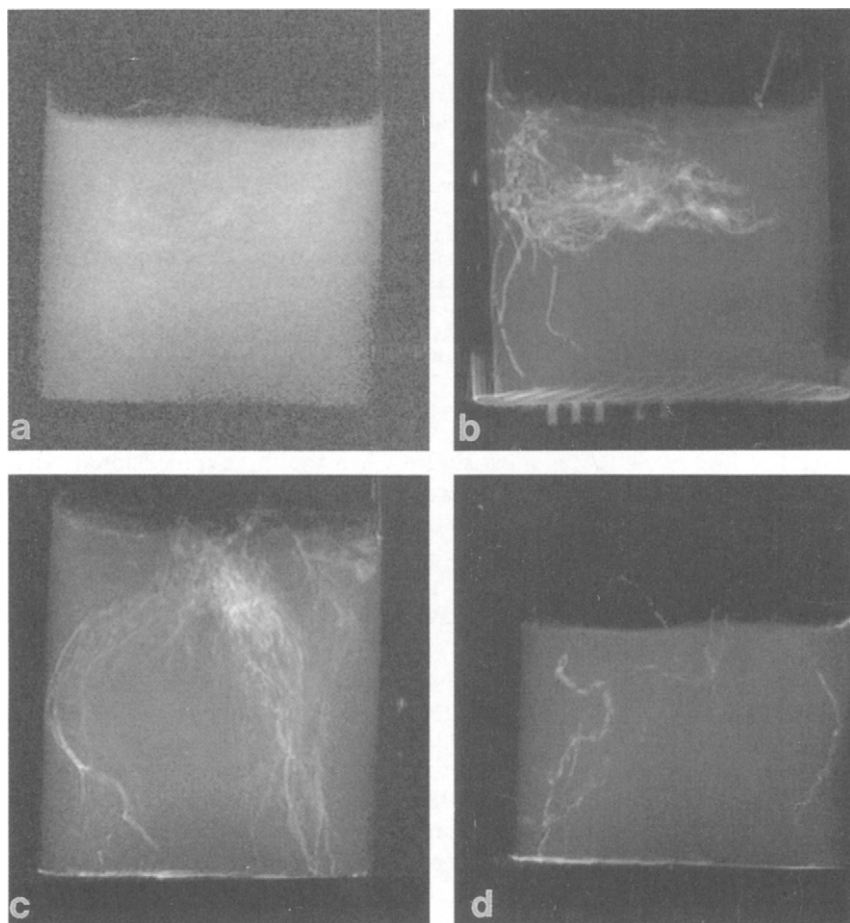


Fig. 10-7. Plant roots in plaster of paris after short wetting periods. (a) bean roots after a 5-min wetting period, (b) bean roots after two 5-min wetting periods, (c) sunflower roots after two, 5-min wetting periods, and (d) six trimmed sunflower roots after two, 5-min wetting periods.

nels, the water conductance of decaying root tissue, and the flow velocity through root channels. Future studies also need to verify whether or not the apparent wicking phenomenon occurs in larger scale root-soil systems, or with a slower water application rate.

Although the spatial characteristics of root channels did not affect the occurrence of preferential flow, the water conductivity of permeating channels over time was expected to be different from that of the nonpermeating channels. Nonpermeating root channels had finite lengths in the porous matrix, and fast channel flow was believed to occur only before the channels were saturated. Thereafter the water flow rate in nonpermeating channels was controlled by the rate of water diffusion into the porous matrix (Smettem, 1986; Germann & Beven, 1985). The average hydraulic conduc-

tivity k_1 of plaster of paris was $\approx 10^{-4} \text{ mm s}^{-1}$ (Table 10-1), which was in the same range as the soils with very slow permeability defined by USDA (Hudson, 1981). After a 4-wk decay period, the average hydraulic conductivity k_2 of the bean sample increased by one order (Table 10-1), and was in the slow permeability range. The increase in the average hydraulic conductivity of the bulk sample was due to preferential flow in permeating roots (Table 10-1). The rate of root decay in natural soils is expected to be higher than that in plaster of paris because of the abundant microorganisms in soils. The occurrence and the rate of preferential water flow, however, may not depend on the decay process alone. Processes such as the development of microbial colonies, the dislocation of fine soil particles, and the shrinking and swelling behaviour of clay should also be evaluated in natural soils (Daniel & Bouma, 1974; Taylor & Jaffé, 1990; Bouma & Wösten, 1979; Beven, 1991).

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